Anish Acharya, PhD

Senior Research Scientist
Amazon Web Services (AWS)

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Website • Scholar • GitHub • LinkedIn

Education

- **The University of Texas at Austin** — **Austin, TX**
  PhD, Electrical and Computer Engineering — *Fall, 2019 – Spring, 2025*
  Advisor(s): Prof. Inderjit S. Dhillon and Prof. Sujay Sanghavi
  PhD Committee: Prof. Alex Dimakis, Prof. Ludwig Schmidt, and Prof. Nhat Ho
  PhD Thesis: Robust and Efficient Learning in High Dimensions from Noisy Data

- **University of California, Irvine** — **Irvine, CA**
  M.S., Electrical Engineering and Computer Science — *Fall, 2013 – Fall, 2014*

- **Jadavpur University** — **Kolkata, India**
  B.E., Electronics and Instrumentation Engineering — *Spring, 2009 – Spring, 2013*
  Advisor(s): Amitava Gupta and Shantanu Das

Research Interests

My recent focus is **Recursive Self Improvement (RSI)**: the loop in which an agentic system proposes changes to its own policy (prompts, tools, orchestration logic and weights), scores them, and keeps what scored better. I work on both halves of that loop, the **signal** it optimizes against and the **search** that exploits it, together with the theory of when iterating provably helps. More broadly: **robust reward modelling and evaluation** under biased or adversarial supervision, **multi-agent systems**, **constrained combinatorial search**, and **robust, distributed and federated optimization**.

Research Experience

- **Amazon** — **Austin, TX**
  Senior Research Scientist, AWS (Full Time) — *Spring, 2025 – Present*
  Building Decentralized AI Marketplace.
  Lead the full life-cycle of AWS **multi-agent orchestration meshes** – architecture, core research, production integration, and eval-driven iteration. Research on **recursive self improvement** – robust reward modeling (**RoPoLL**), prompt optimization(**SPLICE**), and orchestration search (**OrchOpt**) – directly powers: **AI-assisted Product Listing**, AWS Summit NY 2026 (**blog**, **docs**, **video**); **AWS Partner Central Agents** (**onboarding**, **funding**, **MCP**); **AWS Marketplace Agent Mode**, re:Invent 2025 (**blog**, **product**, **ZDNet**, **CRN**, **The Ravit Show**, **Inside Partnering**).

- **University of Texas at Austin** — **Austin, TX**
  Graduate Research Assistant (Part Time) — *Fall, 2019 – Spring, 2025*
  Doctoral research on **robust, sample-efficient learning in high dimensions**: robust estimation under label noise and Byzantine corruption, distributed and federated optimization, and data-efficient subset selection (**thesis**). Affiliated with the **Institute for Foundations of Machine Learning (IFML)**.

- **Eidon AI** — **Austin, TX**
  Head of AI Research (Part Time) — *Summer, 2024 – Winter, 2025*
  The AI is the Blockchain, the Blockchain is the AI.
  Architected and launched a **multi-agent dynamic point-based system** integrating blockchain with AI: evaluation metrics for content quality, AI alignment, and community standards, and a **tokenomics framework for reward distribution** across the decentralized network (**Technical Docs**).

- **Meta** — **Menlo Park, CA**
  Student Researcher (Part Time) — *Fall, 2022*
  Research Scientist (Intern) — *Summer, 2022*
  Student Researcher (Part Time) — *Fall, 2021*
  Research Scientist (Intern) — *Summer, 2021*
  With Meta Capacity Engineering and Meta Fundamental AI Research (FAIR), hosted by **Li Jing**, **Bhargav Bhushanam**, **Dhruv Choudhary**, **and Michael Rabbat**: online, low-resource training to scale Facebook's deep recommendation models. Developed **positive-unlabeled contrastive representation learning** for weakly supervised click-through-rate prediction, learning from positives and unlabeled data without negative sampling – **ICML 2022** PODS workshop (**paper**) and **extended version**.

- **Microsoft** — **Redmond, WA**
  Research Scientist (Intern) — *Summer, 2020*
  As part of the **Speech and Dialog Research Group** (hosted by **Kenichi Kumatani** and **Dimitrios Dimitriadis**), developed algorithms

for training large-scale models robust to **adversarial, Byzantine, and noisy client updates** in the federated setting; built **OpenMSFTL**, an open-source federated learning platform (**github**).

- **Amazon** **Sunnyvale, CA**
  Applied Scientist, AGI (Full Time) *Summer, 2018 – Spring, 2019*
  Applied Scientist, Alexa AI (Full Time) *Spring, 2018 – Summer, 2018*
  Applied Scientist, Alexa ML (Full Time) *Winter, 2017 – Spring, 2018*
  Software Development Engineer, Amazon Devices (Full Time) *Summer, 2016 – Winter, 2017*
  Shipped NLP domain models shared across Alexa devices – Timers, Alarms, Reminders, Calendar, Communications – on Echo Show, Fire TV, and other Alexa IoT devices. **Primary inventor** on: **Alexa Conversations**, goal-oriented dialog generation from dialog templates, APIs, and entity data (**paper**, **patent**, **skill**, **Amazon Science**, **CNN**, **Forbes**, **VentureBeat**); **online embedding compression** for **deployment-constrained NLU** – low-rank factorization folded into training, so task data shapes the compression itself, deployed across a large fleet of per-skill models (**paper**, **patent**, **Amazon Science**, **Amazon Blog**, **Packt**, **Hacker News**).
- **eBay Inc.** **San Jose, CA**
  Data Scientist, Customer Insights and Analytics (Contract – Manthan Systems Inc.) *Fall, 2015 – Summer, 2016*
  Developed **deep recommendation algorithms** to personalize buyer search and targeted advertising, e.g., **generative models for personalized emails**.
- **Schlumberger** **Houston, TX**
  Data Scientist (Full Time) *Spring, 2015 – Fall, 2015*
  Early member of the **Digital Transformation Services** AI research group: **failure prediction** for deployed oil service instruments and **oil level estimation** from extraction life-cycle sensor data.
- **FEM Inc.** (acquired by **Gracenote**, a Nielsen company) **Los Angeles, CA**
  Data Engineer (Full Time) *Winter, 2015*
  With the co-founders: NLP models for a content recommendation system, and visualization tooling for an interactive analytics dashboard.
- **Toyota Technological Institute Chicago (TTIC)** **Chicago, IL; Ann Arbor, MI**
  Visiting Research Scientist (Intern) *Summer, 2014*
  Hosted by **Prof. David McAllester**: real-time stereo algorithm for 3D scene reconstruction for autonomous vehicles, plus a stereo evaluation benchmark based on a slanted-plane scene model; built for **Toyota Research in North America (TRINA)**.

Teaching Experience

- **Amazon Web Services** **Austin, TX**
  Instructor
  AWS GenAI Training: **Embedding AI into Products: Evaluation-Driven Development** *Summer, 2026*
  AWS GenAI Training: **Embedding AI into Products: Governance and Guardrails** *Spring, 2026*
- **University of Texas at Austin** **Austin, TX**
  Instructor *Fall, 2025*
  Directed Reading Program (DiRP): **Natural Language Optimization (NLO) for LLMs**

  Teaching Assistant
  MIS 285N: **Introduction to Deep Learning**, Instructors: Prof. C. Caramanis, Prof. A. Dimakis *Spring, 2024*
  MIS 185N: **Emerging Trends in Machine Learning**, Instructor: Prof. S. Shakkottai *Spring, 2024*
  ECE 461P: **Data Science Principles**, Instructor: Prof. S. Sanghavi *Fall, 2023*
- **University of California Irvine** **Irvine, CA**
  Teaching Assistant
  EECS 50: **Discrete Time Signals and Systems** (Undergraduate) *Fall, 2014*
  EECS 280A: **Advanced Engineering Electromagnetics I** (Graduate) *Spring, 2014*
  EECS 118: **Introduction to Knowledge Management and Software Engineering** (Undergraduate) *Winter, 2014*
  EECS 119: **Introduction to VLSI** (Undergraduate) *Fall, 2013*

Professional Service

- **Reviewer, Conferences:**
  - **International Conference on Machine Learning (ICML)**
  - **Conference on Neural Information Processing Systems (NeurIPS)**
  - **International Conference on Learning Representations (ICLR)**
  - **International Conference on Artificial Intelligence and Statistics (AISTATS)**
  - **Conference on Uncertainty in Artificial Intelligence (UAI)**
  - **Annual Meeting of the Association for Computational Linguistics (ACL)**
  - **IEEE International Symposium on Information Theory (ISIT)**

- **Reviewer, Journals:**
  - **Journal of Machine Learning Research** (MIT Press)
  - **Transactions on Knowledge and Data Engineering** (IEEE)
  - **Pattern Recognition Letters** (Elsevier)
  - **Cognitive Neurodynamics** (Springer Nature)
  - **Discover Electronics** (Springer Nature)
- **Program Committee:**
  - **International Conference on Machine Learning (ICML)**
  - **AAAI Conference on Artificial Intelligence (AAAI)**

Honors and Awards

- **Silver Reviewer** award at ICML 2026. Spring, 2026.
- **Travel Grants**, UT Austin – ML conferences (NeurIPS, ICML, AISTATS, UAI) and Simons Institute, UC Berkeley. 2021 – 2024.
- **Meta PhD Research Grant** of $100,000 Spring, 2021 – Spring, 2023.
- **Meta PhD Research Fellowship Finalist** – AI HW/SW co-design (**Press Release**) Spring, 2022.
- Awarded Bachelor of Engineering with **Highest Honors** (First Class), Jadavpur University Fall, 2013
- **INSPIRE Fellowship** and **MHRD Fellowship for Academic Excellence**, Govt. of India. Fall, 2009 – Spring, 2013.
- **State Rank 5** of ∼700,000 candidates, West Bengal Board of Secondary Education (**The Telegraph**). Spring, 2007
- **Mamraj Agarwal Rashtriya Puraskar** (National Award for Academic Excellence), from the **Governor of West Bengal**. Summer, 2007

Granted US Patents (Primary Inventor)

1. (Amazon) **Goal Oriented Dialog Generation using Dialog Template, API and Entity data**; July, 2022. **USPTO**
   A. Acharya, A. Metallinou, T. Chung, S. Paul, S. Chandra, C. Lin, D. H. Tur, A. Mandal

2. (Amazon) **Natural Language Processing: Amazon Alexa AI**; December, 2020. **USPTO**
   A. Acharya, A. Metallinou, R. Goel, I. Dhillon

Publications

1. **SPLICE: Structured Prompt Local Iterative Combinatorial Evolution**.
   A. Acharya, P. Studans, A. Dhanda, N. V. Rao, V. A. Khatri, S. A. Niaki, B. Verkhovsky.
   Conference on Neural Information Processing Systems (NeurIPS), 2026

2. **RoPoLL: Robust Panel of LLM Judges**.
   A. Acharya, K. Pan, B. Verkhovsky.
   International Conference on Machine Learning (ICML), 2026
   Workshop on Statistical Frameworks for Uncertainty in Agentic Systems (AgenticUQ)
   • OpenReview • Arxiv • Amazon Science

3. **OrchOpt: Validation-Gated Architecture Search for Self-Organizing Multi-Agent Orchestration**.
   A. Dhanda, S. A. Niaki, **A. Acharya**, B. Verkhovsky.
   Conference on Language Modeling (COLM), 2026
   2nd Workshop on Lifelong Agents: Learning, Aligning, and Evolving
   • OpenReview • Amazon Science

4. **Geometric Median Matching for Robust k-Subset Selection from Noisy Data**.
   A. Acharya, S. Sanghavi, A. G. Dimakis, I. S. Dhillon.
   International Conference on Machine Learning (ICML), 2025
   • PMLR • OpenReview • Arxiv

5. **Neural Distributed Source Coding**.
   A. Nagle, J. Whang, **A. Acharya**, H. Kim, A. Dimakis.
   IEEE Journal on Selected Areas in Information Theory, 2024
   • IEEE Xplore • Arxiv

6. **Faster Non-Convex Federated Learning via Global and Local Momentum**. SPOTLIGHT
   R. Das, **A. Acharya**, A. Hashemi, S. Sanghavi, I. Dhillon, U. Topcu.
   The Conference on Uncertainty in Artificial Intelligence (UAI), 2022
   • OpenReview

7. **Robust Training in High Dimensions via Block Coordinate Geometric Median Descent**. SPOTLIGHT
A. Acharya, A. Hashemi, P. Jain, S. Sanghavi, I. Dhillon, U. Topcu.
International Conference on Artificial Intelligence and Statistics (*AISTATS*), *2022*
• PMLR

8. **Positive Unlabeled Contrastive Learning**.
A. Acharya, S. Sanghavi, L. Jing, B. Bhushanam, D. Choudhary, M. Rabbat, I. Dhillon.
International Conference on Machine Learning (*ICML*), *2022*
Principles of Distribution Shift (PODS) Workshop
• Proceedings • Arxiv

9. **LDKP: A Dataset for Identifying Keyphrases from Long Scientific Documents**.
D. Mahata, N. Agarwal, D. Gautam, A. Kumar, S. Parekh, Y. K. Singla, **A. Acharya**, R. R. Shah.
ACM International Conference on Information and Knowledge Management (*CIKM*), *2022*
DL4SR'22: Workshop on Deep Learning for Search and Recommendation
• CEUR-WS • Arxiv • HF Dataset

10. **Alexa Conversations: An Extensible Data-driven Approach for Building Task-oriented Dialogue Systems**. DEMO
A. Acharya, S. Adhikari, S. Agarwal, V. Auvray, N. Belgamwar, A. Biswas, S. Chandra, T. Chung, M. Zarandi, R. Gabriel, S. Gao, R. Goel, D. Hakkani-Tur, J. Jezabek, A. Jha, J. Kao, P. Krishnan, P. Ku, A. Goyal, C. Lin, Q. Liu, A. Mandal, A. Metallinou, V. Naik, Y. Pan, S. Paul, V. Perera, A. Sethi, M. Shen, N. Strom and E. Wang.
Conference of the North American Chapter of the Association for Computational Linguistics (*NAACL*), *2021*
• ACLAnthology

11. **GupShup: An Annotated Corpus for Abstractive Summarization of Open-Domain Code-Switched Conversations**.
L. Mehnaz, D. Mahata, A. Kumar, U. S. Gunturi, R. Jain, G. Gupta, R. Gosangi, I. G. Lee, **A. Acharya**, R. Shah.
Conference on Empirical Methods in Natural Language Processing (*EMNLP*), *2021*
• ACLAnthology

12. **On the Benefits of Multiple Gossip Steps in Communication Constrained Federated Learning**.
A. Hashemi, **A. Acharya**, R. Das, H. Vikalo, S. Sanghavi, I. Dhillon.
IEEE Transactions on Parallel and Distributed Systems, 2021
• IEEE Xplore

13. **Online Embedding Compression for Text Classification using Low Rank Matrix Factorization**. SPOTLIGHT
A. Acharya, R. Goel, A. Metallinou, I. Dhillon.
AAAI Conference on Artificial Intelligence (*AAAI*), *2019*
• Proceedings

14. **Extending The Concept of Analog Butterworth Filter For Fractional Domain**.
A. Acharya, S. Das, I. Pan, S. Das.
Signal Processing (*Elsevier*), *2014*
• ScienceDirect • Arxiv

15. **Simulation studies on the design of optimum PID controllers to suppress chaotic oscillations in a family of Lorenz-like multi-wing attractors**.
S. Das, **A. Acharya**, I. Pan.
Mathematics and Computers in Simulation (*Elsevier*), *2014*
• ScienceDirect • Arxiv

16. **Stability Analysis Of Delayed System Using Bode's Integral**. ORAL
A. Acharya, D. Mitra, K. Halder.
International Conference on Computer Communication and Informatics (*ICCCI*), *2013*
• IEEE Xplore

17. **Optimum PID Control of Multi-wing Attractors in A Family of Lorenz-like Chaotic Systems**. ORAL
A. Acharya, S. Das, I. Pan.
International Conference on Computing, Communication and Networking Technologies (*ICCCNT*), *2012*
• IEEE Xplore

18. **Optimized Quality Factor of Fractional Order Analog Filters with Band-Pass and Band-Stop Characteristics**. ORAL
A. Pakhira, S. Das, **A. Acharya**, I. Pan, S. Saha.
International Conference on Computing, Communication and Networking Technologies (*ICCCNT*), *2012*
• IEEE Xplore

19. **Identification of nonlinear systems from the knowledge around different operating conditions: a feed-forward multilayer Artificial Neural Net (ANN) based approach**. ORAL
S. Saha, S. Das, **A. Acharya**, A. Kumar, S. Mukherjee, I. Pan, A. Gupta.
International Conference on Parallel, Distributed and Grid Computing (PDGC), 2012
- IEEE Xplore

20. **Least square and instrumental variable system identification of AC servo position control system with fractional Gaussian noise**. ORAL
S. Das, A. Kumar, I. Pan, **A. Acharya**, S. Das, A. Gupta.
International Conference on Energy, Automation and Signal (ICEAS), 2011
- IEEE Xplore

Invited Talks and Presentations

- Fall, 2026. (Seattle, WA) **Amazon Machine Learning Conference (AMLC)** – Prompt Optimization (Oral)
- Fall, 2026. (Seattle, WA) **Amazon Machine Learning Conference (AMLC)** – Robust LLM Jury (Oral)
- Summer, 2026. (Seoul, South Korea) **ICML 2026** – Robust LLM Jury (Poster)
- Summer, 2026. (New York City, NY) **AWS Marketplace Science** – Multi-Agent Optimization (Invited Talk)
- Summer, 2026. (Virtual) **AI Across AWS** – Agent Optimization (Invited Talk)
- Spring, 2026. (Virtual) **AWS Town Hall** – Prompt Optimization (Invited Talk)
- Fall, 2025. (Seattle, WA) **Amazon Fulfillment Simulation Symposium (AFSS)** – Robust Sampling (Invited Talk)
- Summer, 2025. (Vancouver, Canada) **ICML 2025** – Robust Sampling (Recorded Talk, Poster)
- Fall, 2023. (Austin, TX) **Future of Chatbots and Conversational AI Summit (FOCA 2023)** (**Keynote**, Panel)
- Fall, 2022. (Virtual) **Meta: Fundamental AI Research (FAIR)** (Invited Talk)
- Fall, 2022. (Berkeley, CA) **Simons Institute for the Theory of Computing** – Robust Optimization (Poster)
- Fall, 2022. (Virtual) **Meta: Ads ML Research** (Invited Talk)
- Summer, 2022. (Eindhoven, Netherlands) **UAI 2022** – Faster Non-Convex Federated Learning (Poster)
- Summer, 2022. (Baltimore, MD) **ICML 2022** – Robust Representation Learning (Recorded Talk, Poster)
- Spring, 2022. (Austin, TX) **6g@UT Forum** – Robust Federated Learning (Poster)
- Spring, 2022. (Virtual) **AISTATS 2022** – Robust Federated Learning (Recorded Talk, Poster)
- Spring, 2022. (Virtual) **University of Exeter – Department of Mathematics** – Robust Federated Learning (Invited Talk)
- Fall, 2021. (Virtual) **University of Denver – Department of Computer Science** – Robust Federated Learning (Guest Lecture)
- Fall, 2021. (Berkeley, CA) **Simons Institute for the Theory of Computing** – Robust Federated Learning (Poster)
- Fall, 2021. (Virtual) **Meta: Fundamental AI Research (FAIR)** – Robust Representation Learning (Invited Talk)
- Summer, 2021. (Virtual) **Meta: Capacity Engineering Research** – Robust Representation Learning (Invited Talk)
- Spring, 2021. (Virtual) **NSF TRIPODS** – Efficient Decentralized Optimization (Poster)
- Summer, 2020. (Virtual) **Microsoft: Azure AI** – Robust Optimization (Invited Talk)
- Fall, 2019. (Austin, TX) **UT Austin Graduate Research Showcase** – Efficient ML (Poster)
- Fall, 2019. (Austin, TX) **Texas Wireless Summit** – Efficient ML (Poster)
- Spring, 2019. (Honolulu, HI) **AAAI 2019** – Efficient ML (Oral)
- Summer, 2018. (Seattle, WA) **Amazon Machine Learning Conference (AMLC)** – Efficient ML (Oral)
- Winter, 2013. (Coimbatore, India) **IEEE ICCCI 2013** – Stability of Delayed Systems (Oral)
- Fall, 2012. (Solan, India) **IEEE PDGC 2012** – Neural Network System Identification (Oral)
- Summer, 2012. (Coimbatore, India) **IEEE ICCCNT 2012** – Chaos Control via Optimal PID (Oral)
- Summer, 2012. (Coimbatore, India) **IEEE ICCCNT 2012** – Fractional Order Analog Filters (Oral)
- Winter, 2011. (Bhubaneswar, India) **IEEE ICEAS 2011** – System Identification under Fractional Gaussian Noise (Oral)
